AMERICAN MARKETING ASSOCIATION

Toward Formalizing Fashion Theory
Author(s): Christopher M. Miller, Shelby H. McIntyre and Murali K. Mantrala
Source: *Journal of Marketing Research*, Vol. 30, No. 2 (May, 1993), pp. 142-157
Published by: American Marketing Association
Stable URL: http://www.jstor.org/stable/3172824
Accessed: 02-01-2018 02:01 UTC

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at http://about.jstor.org/terms

American Marketing Association is collaborating with JSTOR to digitize, preserve and extend access to *Journal of Marketing Research*

JSTOR

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

CHRISTOPHER M. MILLER, SHELBY H. McINTYRE, and MURALI K. MANTRALA*

The authors develop a theoretical framework of the fashion process and a mathematical model of that framework. Basic constructs related to an individual's motivations for adopting fashions are extracted from the fashion literature and related to the individual's fashion decision process. The individual-level model is then integrated into a societal-level framework that can be represented as a system of difference equations. The parameters of the system represent static interpersonal influence networks. The general solution for the system of difference equations is presented and the dynamic implications of several interpersonal influence patterns assumed in previous research are derived mathematically.

Toward Formalizing Fashion Theory

Fashion is important because it is in almost everything.
—Alfred H. Daniels (1951)

Most of society identifies fashion primarily with the field of costume and adornment. However, fashion theorists emphasize that fashion operates in many diverse areas of group life (Robinson 1961). For example, fashion has been suggested to operate in such areas as science (Crane 1969), mortuary practice (Blumer 1969), Christian names (Shepard 1973), and business management (Abrahamson 1991). To limit fashion to the field of costume and adornment is to have an inadequate idea of the true scope of its occurrence (Blumer 1969).

Given the broad presence of fashion's influence, research on the fashion process should be of interest to marketing academicians as well as practitioners. First, fashion phenomena provide a prime example of the social influences that are so often acknowledged to be a prominent aspect of consumer behavior (see Bearden, Netemeyer, and Teel 1989 for a recent treatment). Second, the fashion process is a dynamic example of the use of products to express self-image, role position, or feelings toward relevant others (Belk 1988; Levy 1959). Finally, the fashion process provides an opportunity to study preference shift and formation as opposed to static preferences (Zajonc and Markus 1982).

Despite the importance of fashion and its potential contribution to the marketing discipline, most of the research on fashion has been within the fields of economics, home economics, and sociology. This research provides insightful verbal and anecdotal descriptions of fashion phenomena. However, these descriptions have not yet met the rigorous standards of a consumer theory (Zaltman, LeMasters, and Heffring 1982). A leading author in fashion theory characterizes the situation as follows (Sproles 1981, p. 117):

> The current state of fashion theory includes a loosely organized array of descriptive principles and propositions but is not formalized in that it does not specify a detailed structure of concepts, variables, and relations.

The goal of our article is to advance the understanding of the fashion process, provide relevant managerial insight, and promote future research on this important topic. To this end, we (1) review previous descriptions of fashion phenomena and integrate them into a more formal theoretical framework and (2) develop and analyze a mathematical model of fashion adoptions based on this framework. The mathematical model reveals the type of information needed by managers to forecast fashion trends, in addition to providing a method for analysis and forecast generation. First, we review descriptions of fashion trends and various perspectives of the fashion process that have appeared in the literature. Second, we identify a set of widely accepted constructs from the fashion lit-

*Christopher M. Miller is Assistant Professor of Administrative Science, Jesse H. Jones Graduate School of Administration, Rice University. Shelby H. McIntyre is J. C. Penney Research Professor, Marketing Department, Santa Clara University. Murali K. Mantrala is Assistant Professor of Marketing, College of Business Administration, University of Florida.

The authors express their appreciation to the editor and five anonymous *JMR* reviewers for their valuable suggestions on previous versions of the manuscript.

Journal of Marketing Research
Vol. XXX (May 1993), 142–57

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

erature that influence and motivate an individual's adoption of fashion styles. Third, we develop a representation of the individual's fashion decision process that incorporates the influence of these constructs, and integrate it into a societal-level framework. Fourth, we develop a model of the societal-level framework and examine its dynamic implications in a general sense, then analyze the specific dynamics of social influence patterns assumed by previous fashion theories. We conclude with a summary of the results, implications of our research, and directions for future empirical work and extensions of the theoretical model.

LITERATURE REVIEW

Sproles (1979) has defined a fashion as "a way of behaving that is temporarily adopted by a discernible proportion of members of a social group because that chosen behavior is perceived to be socially appropriate for the time and situation." Fashion is the process of adopting symbols primarily to provide the individual an identity relative to others (Reynolds 1968). Items that serve as socially symbolic for a given situation might include products, product attributes, services, and/or actions. Over time, the meaning of the item, and even which items are deemed socially symbolic, often changes. The fashion process is inherently dynamic and therefore results in a series of interdependent symbolic statements made through time by individuals within a social system.

Though most products, product attributes, services, and actions have utilitarian value to the consumer, the symbolic meaning attached to them provides additional value (Levy 1959; Sirgy 1982; Zaltman and Wallendorf 1979). When items are similar in their utilitarian value, their symbolic value may become a major determinant of choice (Bagozzi 1975; Hirschman 1981; Hirschman and Holbrook 1981; Solomon 1983). In many examples, the symbolic meaning can overcome seemingly large utilitarian deficiencies, as in the cases of tightly laced corsets, high-heeled shoes, and Elizabethan wigs. Therefore, the dynamics of the fashion process can be crucial to the understanding of a consumer's evaluation of products, product attributes, services, and actions within and across times and situations.

Descriptions of Fashion Trends

The literature contains numerous observations and descriptions of fashion trends. These descriptions represent the key phenomena that should be explainable by any conceptual framework and/or mathematical model of the fashion process. This section summarizes the key aspects of those descriptions.

Continuity in fashion trends. The styles adopted within a society as a whole tend to have attributes that are only incrementally different from the attributes of immediately preceding styles (Blumer 1969; Carman 1966; Kroeber 1919; Richardson and Kroeber 1940; Robinson 1958, 1975, 1976; Sapir 1931; Sproles 1981; Weeden 1977; Young 1937). For example, the average skirt length of women's dresses has been found to exhibit relatively continuous cycles in the United States (Richardson and Kroeber 1940).

Cyclical and classic fashion trends. The fashion literature contains descriptions of two main types of fashion trends within a society. One is fashion trends wherein members of a society adopt styles that are progressively more extreme in one direction (e.g., shorter skirt lengths) and then, at later points in time, are progressively more extreme in the opposite direction (e.g., longer skirt lengths). Descriptions of these cyclical fashion trends are given by Barber and Lobel (1952), Blumer (1969), Carman (1966), Grindereng (1967), and Reynolds (1968). However, the cyclical nature of this type of fashion trend is not regular in frequency or amplitude. At times, the "normal" cycle may be interrupted by a midcycle shift in direction (cf. Carman 1966) or the progression may speed up or slow down (Daniels 1951).

The second type of fashion trend described in the literature does not exhibit cyclicality except in the very long run (Sproles 1981; Wasson 1968). This type of fashion trend, termed a "classic," is a convergence within the society on the meaning of a symbolic item that is relatively stable over time. For example, the blue pinstriped suit has been a classic style over a fairly long period.

Speed of fashion trends. Fashion trends may dictate style changes at varying rates across time. Cyclical fashion trends have been described as ranging from extremely fast, sometimes referred to as a "fad," to the extremely slow cycles found for some classics (Meyersohn and Katz 1957; Reynolds 1968; Richardson and Kroeber 1940; Sproles 1981; Wasson 1968).

Fashion turning points. The incremental progression of a cyclical fashion in one direction is most typically continuous until a point of excess or extreme is reached (Robinson 1958). In most cases, the cyclical fashion shifts direction because either (1) the particular style or attribute of a style becomes so extreme that technological constraints preclude further movement in the same direction or (2) the particular style or attribute reaches a point at which strong functional or cultural barriers limit further changes in that direction. For example, during the Middle Ages the Gothic arch became increasingly pointed until the engineering limitations of the time restricted further movement in that direction. Similarly, the late eighteenth century hoopskirt became increasingly wide until navigation became difficult (Robinson 1958). However, in some instances a fashion cycle may shift direction before reaching an extreme (cf. Carman 1966).

Individual-level adherence to societal fashion trends. Though individuals tend to adopt styles in congruence with the fashion trend, each individual appears to behave somewhat randomly in the vicinity around the underlying trend (Blumer 1969; Fromkin 1973; Gross and Osterman 1971). Additionally, a few individuals may adopt styles that are in gross violation of the underlying so-

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

cietal trend during some time periods (Eicher and Kelley 1974).

Fashion Process Perspectives

Diverse and often competing perspectives attempting to explain the fashion process have appeared in the literature. Each of these perspectives adds some insight about the fashion process that we subsequently consider in developing a conceptual framework and mathematical model of the fashion process. In Table 1 we present an adaptation of these perspectives based on the work of Sproles (1985).

We have categorized these perspectives according to two dimensions that summarize the focus of the perspective. The first dimension is titled "internal-external." Perspectives categorized as "internal" view the fashion process as a single, self-contained phenomenon, operating in a diversity of arenas (Robinson 1975), which is primarily driven by endogenous factors within a relatively closed system. Perspectives categorized as "external" view the fashion process as reflecting broader external changes within a society, rarely recognizing commonalities across occurrences of the phenomena (Frings 1991). These perspectives treat the fashion process as primarily driven by exogenous factors. The second dimension, titled "individual-societal," refers to whether the perspective focuses on the fashion process at the level of the individual or the society.

Internal-external. The internal and external perspectives can be viewed as providing alternative approaches to examining fashion phenomena (Robinson 1975). For example, the aesthetic models (cf. Banner 1983; Horn and Gurel 1981) suggest that the clothing fashion process is predominantly a relationship between clothing styles and change in the exogenous societal evaluations of art, ideals of beauty, and/or aesthetic perceptions. Changes in clothing merely reflect changes in these broad, culturally determined values. In contrast, the tricke-down model (e.g., Simmel 1904) suggests that clothing fashion is the result of an endogenous, self-contained process characterized by the sequential adoption of clothing styles by successive social classes. In this model, the fashion process is seen as relatively independent of external forces such as broad cultural changes.

When we compare these two perspectives, it becomes apparent that the external models are very limited in their ability to explain and predict the fashion process. First, the external models are primarily *post hoc* descriptions of the parallels between one exogenous change and a specific shift in fashion (e.g., the sexual revolution and the increased popularity of the miniskirt). As the models do not attempt to explain or predict the specific exogenous change, they are primarily limited to *post hoc* application. For models with the external perspective to be truly explanatory, they would have to include explanations and predictions of the broader exogenous factors and the linkages between those factors and specific types of fashion changes.

Table 1
FASHION PERSPECTIVES
(based on Sproles 1985)

External—Individual Models

—*Aesthetic perceptions and learning model.* The consumer explores new objects, perceives the whole and subcomponents, and forms an initial evaluation. Consumers learn to "like" new stimuli sufficiently different from familiar stimuli, but not those too different or too complex. Later reinforcements increase familiarity and liking. New objects eventually become "acceptable" for fashion status.

External—Social Models

—*Social conflict model.* There can be "anti-fashions" that show stability and cultural identity, and "fashions" that symbolize cultural change. The fashion process is viewed as a competition between these two forces.

—*Art movement model.* Fashions reflect art movements and other cultural changes that are taking place.

—*Ideals of beauty model.* Fashions evolve to follow the changing ideals of beauty held by a society.

—*Mass-market model.* Mass production combined with mass communication makes new styles available simultaneously to all socioeconomic classes. Fashion leaders in each class then adopt new styles at the same time and fashions diffuse through different social classes simultaneously.

—*Market infrastructure model.* Fashion designers, retailers, and the fashion media dictate styles. These businesses control the availability, variety, convenience, and affordability of fashion alternatives.

Internal—Individual Models

—*Demand model.* At higher prices, demand for fashion is limited, but as prices drop demand grows, often substantially. When consumers' tastes are interrelated, the demand curve for fashions may exhibit bandwagon and Veblen effects.

—*Scarcity-rarity model.* Objects that are rare or scarce can have special value and thus command the highest prices.

—*"Conspicuous consumption" model.* The rising wealthy classes desire to display their growing prosperity through conspicuous consumption, which reflects a desire to make the symbolic statement that they are able to afford high priced items.

—*Individualism-centered models.* The accent of contemporary fashion is on the pursuit of individuality, and the importance of being "special" is emphasized in fashion communication.

—*Conformity-centered models.* A vast set of research shows that social pressures to conform can dramatically influence style choices.

—*Uniqueness motivation model.* People compare themselves with others, searching for similarities and differences, and thus form a self-identity. People seek some socially acceptable differences or "uniqueness attributes" in comparison with others, but desire only a moderate amount of such uniqueness.

Internal—Social Models

—*Trickle-down theory.* Once a new object is adopted by the upper class, it is imitated by each succeeding lower class. When each class recognizes that the immediately lower class has become adopters, they move to adopt a new object.

—*Collective behavior theory.* Fashion is viewed as "collective" in that the "fashion industry" suggests fashion possibilities and the collective force of individuals (first retail buyers and later consumers) picks out and converges on what will be "fashionable." The mechanisms of this process are never laid out in explicit detail.

—*Adoption and diffusion model.* Fashion innovators unidirectionally influence fashion imitators toward a particular style. This process continues until all members of a society have adopted the style.

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

Table 1 (Continued)

—*Symbolic communication model.* Styles take on symbolic meanings such as class differentiation and prestige. Styles thus become a visual language.
—*Subculture leadership model.* Subcultures such as black Americans, Native Americans, or the Hell's Angles sometimes prove to be the source of a fashion for another group because of their lower concern for fashion risk.
—*Spatial diffusion model.* Diffusion of innovations occurs as a wave-like process emanating from geographic centers. This model suggests that the infrastructure of the economic system moderates the speed of fashion diffusion.
—*Historical resurrection model.* Designers often look to history for inspirations, and fashion therefore often reflects the past.
—*Historical continuity model.* Each new fashion is an evolutionary outgrowth and elaboration of the previous fashion.

Second, as detailed by Robinson (1975), it seems unlikely that a small set of exogenous factors could be responsible for all fashion phenomena given the diverse domains where the phenomena occur. For example, changes in sexual values may appear to have a causal relationship to fashion trends in skirt lengths, but would not seem to explain the fashion trends found, for instance, in tractor cab design. Thus, a comprehensive theory of the fashion process developed from the external perspective would clearly violate the principle of parsimonious theory construction. Considering these limitations for theory development based on the external perspective, we restrict attention to the internal perspective. We recognize that broader, exogenous changes may influence the fashion process, but they are not the primary drivers of the fashion process.

Individual-societal. The individual-societal dimension refers to whether the focus of the analysis is at the individual or societal level. For example, the individualism-centered models (cf. Fromkin 1973; Gross and Osterman 1971) focus on the motivations of the individual to pursue a unique self-image through the adoption of clothing styles. At the other extreme, the widely accepted collective behavior models (Blumer 1969; Klapp 1969; Lang and Lang 1961; Robinson 1975) suggest that fashion trends result from the collective convergence of a society on what is fashionable.

Upon examining the models along the individual-societal dimension, it becomes clear that, though the levels of analysis differ, the models recognize many of the same aspects of the fashion process. However, the treatment of these aspects varies dramatically along the dimension. Aspects that are subjected to simplification in models at one level of analysis may be the primary focus of models at another level of analysis.

At the individual level, economists have considered bandwagon and snob effects in interdependent demand models (Berg 1973; Granovetter and Soong 1986; Iannaccone 1989; Leibenstein 1950; Stigler and Becker 1977). In these models, the individual-level demand for a product is a function of the aggregate demand of other members of the social system. Higher levels of aggregate demand can increase (decrease) an individual's demand for a product, causing a bandwagon (snob) effect. However, these models seem to have shortcomings when applied to the fashion process because they do not examine the time-dependent spread of demand and they make the simplifying assumption that individuals react only to the undifferentiated aggregate demand of the social system.

At the group level, contagion models of new product adoption have been widely applied to the fashion process (e.g., Baumgarten 1975; Grindereng 1967; King 1963; Schrank and Gilmore 1973; Summers 1970). These fashion-modified adoption and diffusion models describe the time-dependent spread of a single fashion item across a social system, but make severe assumptions about the individuals within that system. For example, the structure of these fashion diffusion models assumes that the social system includes only two homogeneous groups, the innovative fashion leaders and the fashion imitators. Further, it is typically assumed that the innovators are influenced only by external forces, whereas the imitators are influenced only by positive contagion effects such as positive word-of-mouth (see Mahajan, Muller, and Bass 1990). The fashion adoption and diffusion models do not incorporate bidirectional influence and feedback between fashion leaders and fashion imitators (Blumberg 1974; Field 1970) or account for the heterogeneity of selective interaction within the social system (Midgley 1983), or allow for negative contagion effects. Additionally, the fashion diffusion models are limited to examining a single fashion product's diffusion rather than the cyclical adoption of different fashion items, which is the core of the fashion process.

Other group-level models, such as the collective behavior model (Blumer 1969) and the symbolic communication model (King and Summers 1967), emphasize that interpersonal social influence is a two-way process and that the pattern of this influence may vary across individuals within the social system. However, these patterns of social influence have never been explicitly stated, nor have they been related to the dynamic spread of fashions across a social system.

In summary, models across the individual-group dimension of the taxonomy address common aspects of the fashion process and collectively yield substantial insight into that process, and yet no single model seems to capture the richness of both the individual-level and the societal-level perspectives. Therefore, we proceed by presenting the common aspects and insight drawn from both the individual and group models from which our conceptual framework and mathematical model are built.

CONSTRUCTS

There appears to be wide agreement about a limited number of constructs that motivate the fashion decision

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

process at the individual level. To arrive at this conclusion, we listed constructs, their definitions, and how they were used in the fashion literature from such diverse disciplines as sociology, communications, psychology, economics, home economics, and business. We then consolidated the list of constructs by removing duplicates, combining those that differed only semantically, and redefining measurement-oriented constructs in terms of their respective latent equivalents. At the conclusion of this process, we found that the list consisted of only a small number of conceptually distinct and useful constructs that we could use for fashion theory development. These key constructs form the basis of our conceptual framework.

Strength of A Priori Preferences

Fashion theorists recognize that preferences across a range of functionally equivalent styles are weak or nonexistent prior to social interaction (cf. Barber and Lobel 1952; Blumer 1969; Flacus 1906; Sproles 1981). The ability to rank preferences prior to social interaction is particularly low when there is no broadly agreed upon measurement scale for assessing relative performance (e.g., relative performance of a blue sweater vs. a red sweater) or when there is a high degree of uncertainty as to functional performance (e.g., the true efficacy of a management technique; Abrahamson 1991).

Despite weak preferences prior to social interaction, strong preferences can form after social interaction if different symbolic meanings become attached to the styles. For example, a black leather jacket is functionally equivalent to a brown leather jacket and, in the absence of symbolic meaning, individuals might be indifferent between them. However, because of the different symbolic meanings that have evolved, some subcultures strongly prefer the black leather jacket (e.g., the Hell's Angels).

Because symbolic meanings, in general, evolve only in the context of social interaction, preferences between functionally equivalent styles exist only within a specific context and do not exist prior to the consideration of that context. Hence, this construct motivates the individual to continually check his or her social referents to determine the meaning currently being attached to alternative styles.

Though this construct is widely found in theoretical work on fashion, it has only rarely appeared in empirical work on fashion (see Venkatesan 1966 for a notable exception).

Selective Influence

People interact (verbally and nonverbally) at different levels with different individuals while learning the appropriate meaning to attach to alternative styles (Daniels 1951; Midgley 1983; Nystrom 1928; Simmel 1904; Veblen 1912). For example, one individual may interact with and be influenced by a broad-based social circle, whereas another individual may interact with and be influenced by only a few close friends. The social circles of two individuals may or may not overlap. In addition, when overlap is present, the type of influence exerted on the two individuals may differ. A successful businessman may be admired by one individual and influence that individual toward a style, whereas the same businessman may be disdained by a second individual and influence the second individual away from the same style. The differences in breadth and direction of iterpersonal influence across individuals produce the selective influence effect.

We treat the selective influence patterns among individuals within a society as given, but it is clear that the selective influence pattern within a society is itself the product of other social forces and institutions. In particular, the mass media allow certain individuals to exert an extremely broad influence within a society (e.g., Elizabeth Taylor, the editor of Vogue magazine, fashion designers) and thus serve to reduce the selectivity of influences across members of a society. However, even extensive mass media coverage cannot completely homogenize the influence patterns within a society and therefore mass media cannot create individuals with sufficient influence to exclusively dictate fashions trends (Blumer 1969; Daniels 1951).

Selective influence has been one of the most widely used constructs in the fashion literature. For examples of empirical work utilizing this construct in fashion, see Baumgarten (1975), King (1963), Grindereng (1967), Smucker and Creekmore (1972), Clum and Eicher (1972), and King and Summers (1967).

Group Membership/Conformity

Human beings have a natural tendency to form groups. Individuals esteem the members of some groups and shun the members of others. The individual attempts to be perceived by the society as a member of whichever group he or she considers esteemed and attempts not to be perceived as a member of groups that he or she disdains. The fashion literature widely recognizes that adopting the symbols of the esteemed group is one method of communicating membership in that group, whereas avoiding the symbols of disdained groups communicates distance from them. For theoretical treatments, see Leibenstein (1950), Simmel (1904), Veblen (1912), Sapir (1931), Robinson (1961), Dooley (1930), and Jacobson (1936).

As the degree of congruence in the interpretation of these symbolic meanings increases within a society, individuals attempting to be perceived as members of the same group(s) tend toward increased conformity in their adopted symbols. However, as soon as imitators begin to copy the social group's fashion symbol, the fashion symbol no longer serves as a line of demarcation for group membership. Faced with a destruction of the uniformity of the group's coherence, the social group turns away from that fashion symbol and adopts a new one, which reestablishes the demarcation between members and imitators (Simmel 1904).

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

The earliest fashion literature relied on group conformity as the motivator of fashion behavior (e.g., Simmel 1904). Though the literature has broadened in scope to include other constructs, group membership/conformity remains a central construct in both theoretical and empirical work. For examples of the use of this construct in empirical fashion studies, see Gurel, Wilbur, and Gurel (1972), Evans (1964), Smucker and Creekmore (1972), Eicher and Kelley (1974), and Davis and Miller (1983).

Individuality/Differentiation

Within the bounds of group conformity, human beings have a desire to express their individuality through their adopted styles. The desired level of individuality may vary across people and across cultures. For the majority of people, excessive individuality is usually avoided because of the desire for group membership/conformity. For example, when one is attending a party, it is deemed inappropriate to be dressed exactly like someone else, but also inappropriate to appear entirely unique.

In the 1930s, these conflicting forces of group membership/conformity versus individuality/differentiation began to gain more equal attention in the fashion literature, largely because of the inability of previous work to explain fashion behavior adequately on the basis of group conformity alone (Dooley 1930; Jacobson 1936; Sapir 1931). The individuality construct remains one of the most often utilized in theoretical fashion research (see Blumer 1969; Fromkin 1973; Gross and Osterman 1971; Leibenstein 1950; and particularly Klapp 1969), but has been less used in empirical fashion studies (exceptions include studies by Evans 1964 and Gurel, Wilbur, and Gurel 1972).

Desire To Be Current

Though their level of desire may vary, individuals tend to have a desire to adopt styles that will identify them as up-to-date with the people whom they admire in a given situation. Because the meaning attached to different styles may evolve over time, individuals must continually reevaluate the meaning attached to their adopted styles. For example, in the 1950s a white tee-shirt worn with Levis was a symbol of rebellion. However, in the 1960s white tee-shirts with Levis lost this meaning.

This construct has not been clearly defined in most previous literature, but the time-based aspect of conformity appears in both the theoretical and empirical fashion literature (Back 1985; Blumer 1969; Greeno, Sommers, and Kernan 1973; Johnson, Nagasawa, and Peters 1977; Veblen 1912).

Attitude Toward Change

It is widely recognized that people differ on the degree of change that they can tolerate/enjoy. Fashion theorists have noted that some people's self-identity is virtually driven by changing styles and taking fashion risks; however, most people have a strong resistance to changing styles and incurring risks (Blumer 1969; Cobliner 1950; Robinson 1975; Sproles 1981; Taylor 1974; Winakor, Canton, and Wolins 1980). Changing adopted styles involves risk to the individual because a substantial resource investment must be made (e.g., replacing clothing prior to the end of its functionality) and the future meaning, and therefore symbolic utility, is uncertain (Taylor 1974). Additionally, extreme changes in adopted styles may place the individual's consistency of self-image at risk (Sirgy 1982).

A FRAMEWORK FOR UNDERSTANDING THE FASHION PROCESS

The core constructs identified from the literature have an impact on fashions because of their relationship to the individual's fashion decision process. Therefore, we begin by examining the individual's decision process and then integrate that decision process into a framework for societal-level analysis (Figure 1). The individual-level decision process and its relationship to the core constructs are presented as unidirectional for ease of exposition, though we recognize that significant feedback between steps undoubtedly occurs.

Individual Level

An individual searches for information about the symbolic meaning attached to alternative styles from others with whom he or she interacts (Midgley 1983). If *a priori* preferences are strong (perhaps because of large functional differences between alternatives), symbolic motivations will be relatively less important and fashion checking infrequent (Blumer 1969; Sproles 1981). Additionally, if the individual holds a strong desire to be current, checking of the current meaning attached to alternative styles will be more frequent (Back 1985).

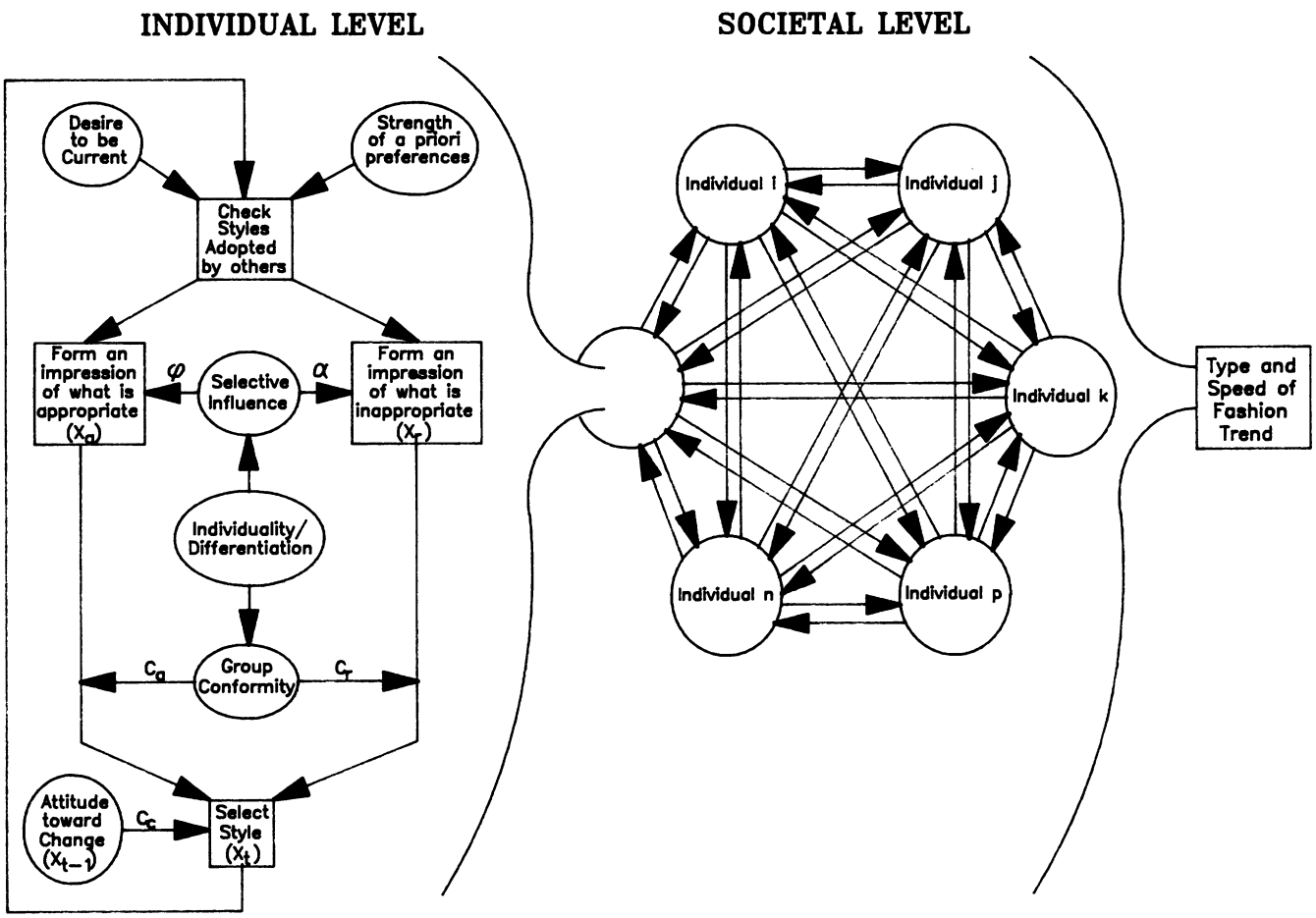
The people referenced by the individual may be held in esteem and influence him or her toward a given symbol or be disdained and influence her or him away from a given symbol (Davis and Miller 1983). For ease of exposition of the conceptual framework (Figure 1) and for reasons detailed in the next section, we separate the two directions of selective influence. The degree of positive selective influence (shown as ϕ in Figure 1) or negative selective influence (α in Figure 1) attributed to each person referenced may vary considerably across individuals.

The influence attributed to others may be different for different domains (Midgley 1983). One individual might be attributed high positive influence as a referent for the symbolic meaning attached to athletic wear, whereas a different individual may be a positive influence referent for determining the meaning attached to alternative business management techniques. The differences in breadth and depth of interaction, and the differences in referencing, are reflections of a person's individuality (Fromkin 1973).

On the basis of this information, the individual forms an impression of which style is appropriate (x_a in Figure

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

Figure 1
THE FASHION PROCESS



1) and which style is inappropriate or repulsive (x_r in Figure 1) for his or her self-identity in a given time and situation. However, this information does not entirely determine the individual's preferences for styles. The individual's attraction to esteemed groups (c_a), desire to be distanced from repulsive groups (c_r), and attitude toward change (c_c) also affect the preference formation process. The level of attraction to esteemed groups (c_a) may not equal the desire of the individual to be distanced from repulsive groups (c_r), which is the motivation to separate the single construct of selective influence into its positive (ϕ) and negative (α) components. The magnitudes of these concerns reflect the individual's susceptibility to interpersonal influence (see Bearden, Netemeyer, and Teel 1989).

Societal Level

The individual decision process is assumed to be followed by all members within the society. However, individuals are heterogeneous with respect to the parameters of the decision process. For example, some individuals have a stronger desire to be current, leading to more frequent checking of the styles adopted by others. Additionally, the same style may hold different meanings across individuals because of differences in referencing and influence.

Because each individual influences and is influenced by other individuals within the society, the society is characterized by interdependencies (Berg 1973). In Figure 1 these interdependencies are shown as arrows between individuals. An individual learns the symbolic meaning(s) attached to a style for a given time and situation by referencing others (inbound arrows), places a value on the symbolic meaning(s), and selects a style to adopt. The individual's adopted style then communicates information to the other members of the social system (outbound arrows). Those members of the social system whom the individual influences may perceive the centers

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

of appropriateness and/or inappropriateness as having changed because of this information and they too will adopt new styles. These style changes then influence others to change styles, including the first individual, and the process continues. In this way, the fashion process is the dynamic, interdependent adoption of items by individuals within the context of a society.

A MODEL OF FASHION ADOPTION DYNAMICS

The fashion process described here is inherently dynamic. In this section we present a mathematical model of the process, which we subsequently analyze to derive the framework's dynamic implications.

Though the fashion process may operate on multiple items and/or multiple attributes of the same item simultaneously, for simplicity we assume individuals choose the level of an attribute x from a continuous one-dimensional space that is constrained at each end by physical limits. Because differences in a single attribute often define different styles (e.g., differences in skirt length define different products in the skirt category), we refer to the choice of a specific x as "selecting a style."

As indicated in Figure 1, the individual's goal is to select x_t that maximizes his or her utility in each time period. For example, the consumer may be choosing a style of tie on the basis of its width in time period 10 and may select a style with a 75-millimeter width as the utility-maximizing style ($x_{10} = 75$). The utility for each style is derived from the individual's evaluation of the meaning attached to that style. However, the meaning attached to the style for a given time period t is based on the position of that style in relation to the styles adopted by other people (e.g., a 75-mm tie width may be interpreted as completely unacceptable if members of the esteemed group are wearing much wider ties). As the individual does not know exactly what styles others will choose in the current period, we assume the individual bases his or her current selection on the styles others were observed adopting in the previous period $t - 1$.

Mathematically, denoting individual i's adoption in period t by x_{it}, we have:

$$x_{it} = f(\mathbf{X}_{t-1}) \tag{1}$$

where $\mathbf{X}_{t-1} = (x_{it-1}, \ldots, x_{nt-1})'$ is the column vector of adoptions in period $t - 1$ by the n members of the society. We now develop an explicit formulation of this general relationship, beginning with a specification of an individual's utility function.

Preference Formation

As stated previously, the individual is selectively influenced by the other members of the social system. Individual i in time period t takes a weighted average, denoted as x_{ait}, of the styles chosen by positive reference individuals in the previous period as the *appropriate* style for the current time and situation. Members for whom the individual consumer has great respect as references in the specific situation will have high weights, with these weights being non-negative and summing to unity. The individual assigns a zero weight to self because selective influence is a socially derived concept. In an analogous way, individuals form an impression of what is *inappropriate* by utilizing the relative weight of social system members who are disdained. In this second weighted average (x_{rit}), individuals who are highly disrespected will have high weights, again non-negative and summing to unity. Separating the selective influence construct into two vectors allows for the representation of potentially different weights assigned to the level of attraction to esteemed individuals and the level of repulsion from disdained individuals.

Thus, for all $i = 1, \ldots, n$,

$$x_{ait} = \boldsymbol{\phi}_i \mathbf{X}_{t-1} \tag{2}$$

$$x_{rit} = \boldsymbol{\alpha}_i \mathbf{X}_{t-1} \tag{3}$$

where:

- $\boldsymbol{\phi}_i$ is a row vector of *respect* weights assigned by individual i to others and
- $\boldsymbol{\alpha}_i$ is a row vector of *disrespect* weights assigned by individual i to others.

Symbolic Utility

Upon determining which styles are appropriate and inappropriate for self-identity, the individual is assumed to adopt the style that maximizes his or her utility. Specifically, we assume the utility of a style is an additively separable function of (1) how close the style is to what is perceived as appropriate, (2) how far the style is from what is perceived as inappropriate (or repulsive), and (3) the distance the style is from the individual's last adopted style. As in the standard ideal point utility models that have appeared in the consumer literature (cf. Carroll 1972; McAlister and Pessemier 1982), we assume a simple quadratic specification of the components of the utility function, namely,

$$U_i(x_{it}) = -c_{ai}(x_{it} - x_{ait})^2 + c_{ri}(x_{it} - x_{rit})^2 - c_{ci}(x_{it} - x_{it-1})^2 \tag{4}$$

where:

- x_{it} is the style chosen by individual i in period t,
- $U_i(x_{it})$ is individual i's utility function, and
- c_{ai}, c_{ri}, c_{ci} are positive parameters indicating individual i's concerns for appropriateness, inappropriateness, and change, respectively.

Given this utility function, the first-order condition for utility maximization is

$$-c_{ai}(x_{it} - x_{ait}) + c_{ri}(x_{it} - x_{rit}) - c_{ci}(x_{it} - x_{it-1}) = 0. \tag{5}$$

Note that the second-order condition to ensure utility maximization rather than minimization is given by

$$-c_{ai} + c_{ri} - c_{ci} < 0. \tag{6}$$

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

Solving equation 5 for the utility maximizing x_{it}^*, we obtain

$$x_{it}^* = \frac{-c_{ai}x_{ait} + c_{ri}x_{rit} - c_{ci}x_{it-1}}{-c_{ai} + c_{ri} - c_{ci}}. \quad (7)$$

Substituting equations 2 and 3 into equation 7, we obtain

$$x_{it}^* = \frac{-c_{ai}(\phi_i \mathbf{X}_{t-1}) + c_{ri}(\alpha_i \mathbf{X}_{t-1}) - c_{ci}x_{it-1}}{-c_{ai} + c_{ri} - c_{ci}}. \quad (8)$$

Equation 8 is individual i's utility-maximizing style in period t, for $i = 1, 2, \ldots, n$.

Considering all individuals simultaneously, it follows that

$$\mathbf{X}_t = \boldsymbol{\Omega}\, \mathbf{X}_{t-1}, \quad (9)$$

where $\boldsymbol{\Omega}$ is a summary square matrix of social influence patterns whose main diagonal elements $\{a_{ii}\}$ are given by

$$a_{ii} = \frac{-c_{ci}}{-c_{ai} + c_{ri} - c_{ci}} \qquad \text{for } i = 1, \ldots, n, \quad (10)$$

and represent the ith individual's concern for change relative to all three concerns. Recalling the second-order condition for utility maximization (equation 6) and that the concern parameter c_{ci} is defined to be positive, we see that all diagonal elements of $\boldsymbol{\Omega}$ are positive.

The off-diagonal elements of $\boldsymbol{\Omega}$ are given by

$$a_{ij} = \frac{-c_{ai}\,\phi_{ij} + c_{ri}\,\alpha_{ij}}{-c_{ai} + c_{ri} - c_{ci}} \qquad \text{for } i \neq j, \quad (11)$$

where ϕ_{ij} and α_{ij} are the respect and disrespect weights assigned by individual i to individual j. The off-diagonal elements represent the net influence of the jth consumer on the ith consumer, where the denominator simply normalizes this net influence by the three concerns of the ith consumer. A value of a_{ij} near zero means the ith consumer gives little weight to the jth consumer's adopted style when selecting his or her own. A positive (negative) value implies the ith consumer wants to adopt a style similar (dissimilar) to the adopted style of the jth consumer. The relative magnitudes of the diagonal elements a_{ii} and the off-diagonal elements a_{ij} can be interpreted as the susceptibility to interpersonal influence of the ith consumer.

It is important to note that the sum across each row of the $\boldsymbol{\Omega}$ matrix equals 1.0 because the elements in any given row represent all of the relative influences on a single individual, including the individual's self-influence from the past period. However, the sum down each column represents the level of influence a given individual has on the rest of the social system and therefore will most likely not sum to one. A broadly respected individual would have a high column sum, whereas a broadly disrespected individual would have a negative column sum. The difference in column weights across individuals can be interpreted as the level of social discrimination in the social setting.

Further analysis is required to examine the fashion model's ability to yield the full diversity of fashion results. In the next section, we first outline the general mathematical solution for individual time paths resulting from any $\boldsymbol{\Omega}$ matrix and then investigate the nature of the time paths resulting from several specific $\boldsymbol{\Omega}$ matrix patterns suggested by previous fashion theories.

General Model Solution

Equations 9, 10, and 11 describe a first-order, homogeneous system of n difference equations (see Chiang 1984; Gandolfo 1980). A general mathematical solution to the system of difference equations consists of determining an expression for the fashion time paths, that is, the time paths of style choices by specific individuals.

The solution approach involves first considering trial solutions of the type $x_{it} = \gamma_i \lambda^t$ where γ_i, $i = 1 \ldots n$, are arbitrary constants (not all zero) and the base λ is a nonzero parameter (Chiang 1984). On the basis of equation 9, the trial functions will be a solution of the system for any t if and only if

$$(\boldsymbol{\Omega} - \lambda \mathbf{I})\boldsymbol{\gamma} = \mathbf{0}, \quad (12)$$

where $\mathbf{I}$ is the identity matrix of order n and $\mathbf{0}$ is a column vector of zeros. A necessary and sufficient condition for determining the particular values for λ for which there is a nonzero vector $\boldsymbol{\gamma}$ is that the determinant of $(\boldsymbol{\Omega} - \lambda \mathbf{I})$ be zero.

Using the diagonal expansion of the determinant, we find the characteristic equation of $\boldsymbol{\Omega}$ to be

$$\sum_{k=0}^{n} (-\lambda)^{n-k}\, tr_k(\boldsymbol{\Omega}) = 0, \quad (13)$$

where $tr_k(\boldsymbol{\Omega})$ is the sum of the principal minors of order k of $|\boldsymbol{\Omega}|$, and we define $tr_0(\boldsymbol{\Omega}) = 1$ and $tr_n(\boldsymbol{\Omega}) = |\boldsymbol{\Omega}|$. Thus $tr_1(\boldsymbol{\Omega})$ is the principal diagonal trace of $\boldsymbol{\Omega}$.

The characteristic equation 13 has exactly n roots $\lambda_1, \lambda_2, \ldots, \lambda_n$. Once the roots are determined, one can compute the corresponding nonzero vectors $\boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2, \ldots, \boldsymbol{\gamma}_n$. As all the characteristic roots affect the time paths, the general solution for the ith time path is a linear combination of the n individual solutions, that is, for $i = 1, \ldots, n$,

$$x_{it} = \sum_{l=1}^{n} A_{il} \lambda_l^t \quad (14)$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the roots of the characteristic equation of $\boldsymbol{\Omega}$ and $A_{i1}, A_{i2}, \ldots, A_{in}$ are constants that are assigned definite values with the help of initial conditions.

As seen in equation 14, the time path of style choices resulting from any $\boldsymbol{\Omega}$ matrix depends on its characteristic roots, which may be *real* or *complex*, *positive* or *negative*, and may vary in magnitude. We now discuss the implications of each type of characteristic root.

Positive, real roots. The time paths of fashion adoptions within this type of society will be monotonic over

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

time. If the dominant root λ_d (i.e., the characteristic root with the largest magnitude) is greater than 1.0, individuals within the society adopt increasingly extreme styles over time as the term λ_d^t grows exponentially as t increases. This process would continue until the physical or practical limits of the style are approached. As an individual approaches this limit, he or she may discover that the highest utility adoption is to step in the opposite direction rather than continuing to move incrementally toward the limit (cf. Hicks 1950). Therefore, the physical limits to the style redirect the monotonically diverging time path of the fashion adoptions in the opposite direction, producing over time a cyclical fashion ranging from one limit extreme to the other. We term this type of fashion a *constraint-induced cyclical fashion.*

However, if the dominant root is less than or equal to 1.0, each individual will monotonically tend toward a style that becomes constant over time. The style that each individual tends toward may or may not be different across individuals. When all individuals move toward adopting the same style, we term this type of fashion a *classic fashion.*

Negative real roots. If all the characteristic roots of an Ω matrix are real but some are negative, the time path of fashion adoptions within that society will display discontinuous fluctuations as the negative roots fluctuate in sign with t even versus odd. These sign fluctuations on the negative real roots, when added to the other roots of the characteristic equation, produce discontinuous fluctuations in the time path. Hence, the time paths do not appear "smooth" when some negative roots are present. If the negative roots are greater (less) than 1.0 in magnitude, these fluctuations increase (decrease) in magnitude over time.

Complex roots. Finally, if some of the characteristic roots of an Ω matrix are complex, the time path of fashion adoptions within that society will display continuous oscillations. Complex roots always occur in pairs having the form $\alpha \pm \beta i$ where $i = \sqrt{-1}$. The nature of the oscillations due to complex roots can best be seen by considering an Ω matrix in which the kth and $k + 1^{\text{st}}$ roots are a complex pair, such as

$$\lambda_1, \lambda_2, \ldots, \lambda_{k-1}, \alpha + \beta i, \alpha - \beta i, \lambda_{k+2} \ldots, \lambda_n,$$

which results in the time path of x_{it} being represented as

$$x_{it} = A_1\lambda_1^t + A_2\lambda_2^t + \ldots + A_k(\alpha + \beta i)^t + A_{k+1}(\alpha - \beta i)^t + \ldots + A_n\lambda_n t. \tag{15}$$

This expression can be equivalently represented in polar coordinate form as

$$x_{it} = A_1\lambda_1^t + A_2\lambda_2^t + \ldots + r^t \cdot (A_k' \cos \omega t + A_{k+1}' \sin \omega t) + \ldots + A_n\lambda_n t \tag{16}$$

where A_k', A_{k+1}', r, and ω are the constants associated with the complex roots. If the complex roots are dominant, the resulting trigonometric oscillations will increase in amplitude over time if $|r| > 1.0$, decrease in amplitude if $|r| < 1.0$, and be of constant amplitude if $|r| = 1$. The presence of complex roots resulting in oscillatory time paths of fashion adoptions is of particular interest because they yield a smooth cyclical time path where the fashion cycle reverses itself prior to reaching the physical limits of the style. We term this type of fashion a *nonconstrained cyclical fashion.*

Given the general mathematical solution for individual time paths resulting from any Ω matrix, we now can investigate the nature of the time paths resulting from several specific Ω matrix patterns suggested by previous fashion theories.

SOLUTIONS OF CONSTRAINED Ω PATTERNS

In the preceding model, the Ω matrix has the following basic properties.

—Property 1. Ω is a square matrix.
—Property 2. Ω contains only real elements.
—Property 3. The elements of each row of Ω sum to unity, that is,

$$\sum_{j=1}^{n} a_{ij} = 1, \qquad \forall i = 1, 2, \ldots, n.$$

—Property 4. All elements along the leading diagonal of Ω are strictly positive, that is

$$a_{ii} > 0, \qquad \forall i = 1, 2, \ldots, n.$$

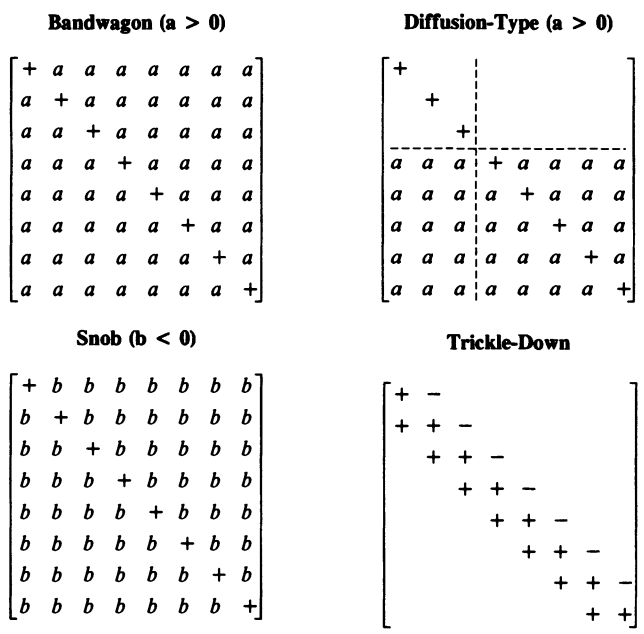
Given these properties, we now derive results for the fashion time paths for some special Ω matrix patterns. Specifically, we analyze four patterns of social influence assumed by previous fashion models as displayed in Figure 2. The objectives of this analysis are (1) to examine whether these models can explain the full range of fashion trends described in the literature and therefore can be considered acceptable general representations of fashion dynamics, (2) to gain insight into the time paths that would result if the assumed pattern of social influence were actually present within a society, and (3) to demonstrate how the proposed model can be used to investigate the fashion time paths resulting from any specified pattern of social influence.

Bandwagon Model

The bandwagon model (e.g, Veblen 1912) assumes individuals are positively influenced by the undifferentiated aggregate demand for an item. Given that the bandwagon model does not recognize differences in individual influences, one reasonable representation of bandwagon effects at the individual level is to treat the individuals as homogeneous, each having an equal, positive influence on everyone else (top left of Figure 2). With this representation, the bandwagon Ω matrix is symmetric with real, positive, and equal off-diagonal elements. Therefore, we can immediately rule out complex roots because a real symmetric matrix has only real roots. The absence of complex roots implies that the bandwagon model cannot yield the continuous cyclical

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

Figure 2
Ω MATRIX PATTERNS OF PREVIOUS MODELS



time paths of fashion adoptions found in the descriptive literature.

Furthermore, recalling basic properties 3 and 4 of the Ω matrix, we note that all the diagonal elements of a bandwagon Ω matrix must be equal and all elements of the bandwagon matrix must have values between zero and one. Therefore, the bandwagon Ω matrix is a "stochastic" matrix whose characteristic equation's dominant root is unity (Gantmacher 1959). Because the bandwagon model's dominant root is not greater than unity, the time paths of fashion adoptions cannot diverge, ruling out the constraint-induced cyclical fashions described in previous literature.

Finally, if all members of the society have self-influence that is greater than the interpersonal influence from any other individual, that is, if the bandwagon Ω matrix has a dominant diagonal, the presence of negative roots can be eliminated. Let the elements of Ω be $a_{ii} = d$ and $a_{ij} = a$ for $i = 1, 2, \ldots, n$; $j = 1, 2, \ldots, n$. Given a dominant diagonal, $d > a$. Let $d - a = b > 0$. Then,

$$\Omega = \begin{bmatrix} a & a & \ldots & a \\ a & a & \ldots & a \\ \vdots & \vdots & \ldots & \vdots \\ a & a & \ldots & a \end{bmatrix} + \begin{bmatrix} b & 0 & \ldots & 0 \\ 0 & b & \ldots & 0 \\ \vdots & \vdots & \ldots & \vdots \\ 0 & 0 & \ldots & b \end{bmatrix} = A + B \tag{17}$$

and

$$|\Omega| = |A + B| = \sum_{k=0}^{n} b^{n-k}\, tr_k(A). \tag{18}$$

In equation 18, it is apparent that $|A| = tr_n(A)$, as well as all its minors of order 2 or more, are zero, that is, $tr_2(A) = tr_3(A) = \ldots = tr_{n-1}(A) = 0$. Therefore,

$$|\Omega| = b^n + b^{n-1} tr_1(A) = b^n + b^{n-1}\, na, \tag{19}$$

which is positive because $a > 0$ and $b > 0$. In a similar manner, all the principal minors of order k of $|\Omega|$, $k = 0, 1, \ldots, n - 1$, are positive. This implies that with a dominant diagonal, the bandwagon Ω matrix is a positive definite real symmetric matrix. Such matrices have only positive roots. Therefore, when a bandwagon model is used to represent a society in which individuals have self-influence greater than the interpersonal influence from any other single individual, the model allows for only smooth time paths, ruling out the fluctuating fashion adoption time paths described in the literature.

In summary, the bandwagon Ω matrix has real, positive roots with the dominant root equal to unity. Therefore, on the basis of the general model solution discussed previously, a society characterized by this pattern of social influence will not experience continuous cyclical fashion time paths, constraint-induced fashion cycles, or, with a dominant diagonal, fluctuations in the fashion time paths. Because time paths from this model can only be monotonic and can only be classic fashions, the bandwagon model is an inadequate general representation of the fashion process because it precludes all other types of fashion trends described in the literature.

Diffusion-Type Models

Diffusion-type models of fashion propagation typically assume two homogeneous classes of individuals described as innovators and imitators (e.g., Grindereng 1967). The pattern of social influence assumed in the mathematics of these models can be represented by the partitioned, non-negative Ω matrix shown in the top right of Figure 2. Specifically,

$$\Omega = \begin{bmatrix} D_1 & 0 \\ D_2 & D_3 \end{bmatrix}, \tag{20}$$

where D_1 is a square diagonal matrix reflecting the independence of innovators from the influence of all other individuals. The lower two partitions, D_2 and D_3, reflect an assumed pattern of imitators being influenced only positively and in equal amounts.

As with the bandwagon model, suppose an imitator's self-influence is greater than the influence from any other individual, that is, D_3 has a dominant diagonal. Considering the characteristic equation of the diffusion-type Ω matrix, note that

$$|\Omega - \lambda I| = |D_1 - \lambda I||D_3 - \lambda I|, \tag{21}$$

which implies the eigenvalues of Ω are the eigenvalues of D_1 combined with the eigenvalues of D_3. The eigenvalues of the diagonal submatrix D_1 are its diagonal elements, which by basic property 3 must all equal unity. Further, the submatrix D_3 is a bandwagon matrix, which

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

has been shown to have only positive roots with the dominant root equal to unity. Therefore, the diffusion-type Ω matrix has only positive roots with the (repeated) dominant root equal to unity, implying that societies characterized by this pattern of social influence will result in imitators that display monotonic, classic fashions (innovators remain at their initial styles). Like the bandwagon model, the diffusion-type model is therefore an inadequate general representation of the fashion process because it precludes all other types of fashion trends described in the literature.

Snob Model

The snob model (e.g., Veblen 1912) assumes individuals are negatively influenced by the undifferentiated aggregate demand for a style. As in the bandwagon model, we represent this at the individual level by treating all individuals as homogeneous, each having an equal and negative influence on everyone else (bottom left of Figure 2).

By basic property 3 of Ω, if the off-diagonal elements of Ω are all negative, the principal diagonal elements of Ω are greater than one. The implication is that the magnitude of the dominant root $|\lambda_d|$ of the snob Ω matrix is greater than one. We prove this by contradiction.

Let λ_1 be the dominant root and suppose $|\lambda_1| < \min a_{ii}$. It follows that

$$\min a_{ii} > |\lambda_1| \geq |\lambda_2| \geq \ldots \geq |\lambda_n|, \tag{22}$$

that is,

$$n(\min a_{ii}) > \sum_{i=1}^{n} |\lambda_i| \geq \sum_{i=1}^{n} \lambda_i. \tag{23}$$

However, because the sum of the eigenvalues of any matrix equals its trace, we obtain

$$\sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} a_{ii} \geq n(\min a_{ii}), \tag{24}$$

which contradicts equation 23. Therefore, $|\lambda_1| \geq \min a_{ii}$ and, as the diagonal elements of the snob Ω matrix are all greater than one, it follows that the magnitude of the dominant root must be greater than one. Therefore, the snob pattern of interpersonal influence rules out classic fashions.

Furthermore, because the snob Ω matrix is a real symmetric matrix, it cannot have complex roots. Hence, continuous fashion cycles are ruled out by this model. Additionally, if the magnitude of the off-diagonal elements is less than or equal to one (i.e., when interpersonal influence is relatively weak), this pattern of social influence cannot have negative roots (proof available from the authors). In such a case, the snob model precludes the fluctuations in the fashion adoption time paths found in the description literature.

In summary, societies characterized by the snob model presented in Figure 2 will not have continuous fashion cycles or classic fashion trends. The model does permit constraint-induced fashion cycles, but it is clearly not an adequate general representation of the fashion process.

Trickle-Down Theory

The trickle-down theory of fashion propagation (e.g., Simmel 1904) posits a social influence pattern in which each social class is influenced toward the style adopted by the immediately higher social class and influenced away from the style adopted by the immediately lower social class. A strict interpretation of the trickle-down theory suggests a "tridiagonal" Ω matrix with zero elements everywhere but on the principal diagonal (positive elements), on the diagonal just above the principal diagonal (negative elements), and on the diagonal just below the main diagonal (positive elements). In this highly restrictive pattern of social influence, shown at the bottom right of Figure 2, individuals are ordered by social strata from the uppermost class, $i = 1$, to the lowest class, $i = n$.

The diagonal expansion of the trickle-down Ω matrix reveals that

$$tr_k(\Omega) > 0 \qquad \forall\, k = 0, 1, \ldots, n. \tag{25}$$

This means the coefficients of the characteristic equation of Ω will alternate in sign. Therefore, by Descarte's theorem, negative roots will not exist (Gandolfo 1980) and discontinuous fluctuations in the fashion trend are ruled out. However, as the matrix is not symmetric, one cannot rule out complex roots. Further, applying basic property 3, we see that at least one of the diagonal elements is less than unity. Therefore, it is not possible to determine whether the dominant root has a magnitude less than or greater than unity without more conditions being imposed. The implication is that societies characterized by the trickle-down pattern of social influence could display all types of fashion trends with the exception of fluctuations. These results substantiate the persistent appeal of the trickle-down model as a general rationale for fashion propagation, despite critiques about the restrictive nature of the assumed social influence pattern (cf. Blumer 1969).

To summarize, the analyses reveal two primary benefits of the proposed model. First, the proposed model is a more general representation of the fashion process than previous models. In fact, each of the four well-known models is a special case of the proposed model, obtained by imposing constraints on the pattern of interpersonal influence. Second, the general solution of the proposed model provides a method by which any specific pattern of social influence can be analyzed. Using this method, we found that the previous models cannot explain all of the fashion trend descriptions found in the literature. In contrast, the proposed model allows for unrestricted patterns of social influence and is capable of producing the full range of fashion trends described in the literature.

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

SUMMARY, IMPLICATIONS, AND FUTURE RESEARCH

The fashion process is an important market phenomenon and determines the fortune of many companies and industries. However, the conceptualization of the fashion process and the examination of its dynamic nature have been underresearched areas in the marketing discipline. After reviewing fashion phenomena and categorizing models that have attempted to explain them, we drew key individual-level constructs and their relationships from previous models and integrated them into a conceptual framework. A system of difference equations to model fashion adoptions was then developed on the basis of the framework.

The model is shown to be capable of explaining the full range of fashion trends described in the literature. The patterns of interpersonal influence assumed by four previous fashion models are restricted special cases of the proposed model. Each is found to be inadequate for explaining the full range of fashion trends. The analyses illustrate how the proposed model can be used to predict fashion dynamics from any specified pattern of social influence within a society.

More generally, our research demonstrates that a deterministic model can endogenously explain the full range of fashion time paths, including fashion cycles and fluctuations. This point is important as fashion theorists have attempted to explain these "aberrations" by various exogenous random shocks or forces (cf. the aesthetic models of Banner 1983; Horn and Gurel 1981). Clearly, it is more theoretically appealing and managerially useful to have a deterministic model of the fashion process that is capable of generating the full range of fashion time paths.

Furthermore, the proposed model examines successive product adoptions from a perspective that is entirely different from the perspective taken in recent extensions of the diffusion model (cf. Norton and Bass 1987). The focus of those models has been the successive adoption of products based on their functional superiority over preceding versions (e.g., the adoption of portable computers based on 286, 386, and 486 microprocessors). Each successive generation ultimately dominates the preceding generation because product substitution is restricted to be unidirectional toward higher functionality. In contrast, the proposed model examines the adoption process across products that are *not* demonstrably superior to each other and product substitution is *not* restricted to be unidirectional. The implication is that the proposed model may be more applicable than these diffusion model extensions for examining successive product adoptions in product-markets wherein competing brands are near functional substitutes.

Clearly, the next step in understanding the fashion process and aiding managerial decisions about products that contain a fashion component is to examine the proposed model empirically. This empirical examination requires more than an examination of past product sales. Empirical work must estimate the social influence pattern between consumers and collect data on *which* consumers are purchasing *which* products. To ease data collection without excessive loss of power, relatively homogeneous market segments could be treated as "individuals" and the model analyzed at the segment level.

Over the years, and in a variety of areas, researchers have assessed interpersonal influence patterns by either direct, cross-sectional measurement (e.g., Festinger, Schacter, and Back 1950) or indirectly inferring the influence pattern on the basis of time series data (e.g., McIntyre and Miller 1992). The following examples show how these approaches might be used to empirically examine the proposed model.

An example of the direct approach using a pre-existing segmentation scheme would be to intercept a cross section of shoppers purchasing a product from the product class of interest. Using the shoppers' responses to segmentation variables developed *a priori*, the analyst categorizes each shopper into one of the pre-existing segments. Next, the shopper's attitudes toward and susceptibility to influence from members of all segments must be assessed on a validated scale of questions. Additionally, the shopper's product purchase(s) should be recorded. Intercepts continue until adequate representation of each segment is obtained. Each element within a row of the Ω matrix is calculated by using a summary statistic of the segment's attitudinal responses and susceptibility to influence responses for each other segment, where the summary statistic is constrained to sum to one across the row. This directly measured Ω matrix from cross-sectional data across *a priori* segments, and the product purchase data from those segments, are then used to forecast purchase trends and/or examine the validity of the model.

An example of the indirect approach, which does not rely on a pre-existing segmentation scheme, would begin by gathering time series data on purchases from a product class for a broad spectrum of consumers and data on the characteristics of those consumers. These purchase time series and consumer characteristics may already be within an organization's database from credit card application and credit card usage records. The analyst uses a standard segmentation method, such as cluster analysis, to construct segments based on the consumer characteristics data. Each segment's purchases during a meaningful unit of time (e.g., season) is regressed on the purchases of all segments in the previous time period to estimate the values for each row of the Ω matrix. The regression coefficients are constrained so that the values for each row of the estimated Ω matrix sum to one. This indirectly measured Ω matrix, together with the latest product purchase data, is used to generate forecasts of future purchase trends and/or empirically validate the proposed model.

Beyond the empirical examination of the proposed model, the framework we propose and the mathematical model derived from that framework both predominantly

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

consider demand-side aspects of the fashion process. Future research should extend this conceptual and mathematical treatment to include key supply-side concepts. For example, the framework should be expanded to include such items as (1) the role of prices and income distributions, (2) an explicit consideration of mass media's impact on the pattern of interpersonal influence, and (3) the role of fashion designers, manufacturers, and retailers in the fashion process.

Furthermore, the proposed model should be extended to consider the fashion process operating on multiple product attributes simultaneously. An obvious extension would be to simply redefine the $\mathbf{X}_t$ vector as a matrix whose rows are consumers (or consumer segments) and whose columns are product attributes (e.g., skirt width, skirt length, neckline). However, this multiattribute extension represents the fashion process as operating on each attribute independently, a restriction that may hamper explanation and prediction of fashion dynamics in some situations. Research is needed to include the potential attribute interrelationships in a multiattribute model of the fashion process.

REFERENCES

Abrahamson, Eric (1991), "Managerial Fads and Fashions: The Diffusion and Rejection of Innovations," *Academy of Management Review*, 16 (3), 586–612.

Back, Kurt W. (1985), "Modernism and Fashion: A Social Psychological Interpretation," in *The Psychology of Fashion*, Michael R. Solomon, ed. Lexington, MA: Lexington Books.

Bagozzi, Richard P. (1975), "Marketing as Exchange," *Journal of Marketing*, 39 (October), 32–9.

Banner, Lois W. (1983), *American Beauty*. New York: Alfred A. Knopf.

Barber, Bernard and Lyle S. Lobel (1952), "Fashion in Women's Clothing and the American Social System," *Social Forces*, 31 (December), 124–31.

Baumgarten, Steven A. (1975), "The Innovative Communicator in the Diffusion Process," *Journal of Marketing Research*, 12 (February), 12–18.

Bearden, William O., Richard G. Netemeyer, and Jesse E. Teel (1989), "Measurement of Consumer Susceptibility to Interpersonal Influence," *Journal of Consumer Research*, 15 (March), 473–81.

Belk, Russell W. (1988), "Possessions and the Extended Self," *Journal of Consumer Research*, 15 (September), 139–68.

Berg, Stanford V. (1973), "Independent Tastes and Fashion Behavior," *Quarterly Journal of Economics and Business*, 13, 49–58.

Blumberg, Paul (1974), "The Decline and Fall of the Status Symbol: Some Thoughts on Status in a Post-Industrial Society," *Social Problems*, 21 (April), 480–98.

Blumer, Herbert (1969), "Fashion: From Class Differentiation to Collective Selection," *Sociological Quarterly*, 10 (Summer), 275–91.

Carman, James M. (1966), "The Fate of Fashion Cycles in Our Modern Society," in *Science, Technology, and Marketing*, Raymond M. Haas, ed. Chicago: American Marketing Association, 722–37.

Carroll, J. Douglas (1972), "Individual Differences and Multidimensional Scaling," in *Multidimensional Scaling*, Roger N. Shepard, A. Kimball Romney, and Sara Beth Nerlove, eds. New York: Seminar Press.

Chiang, Alpha C. (1984), *Fundamental Methods of Mathematical Economics*. New York: McGraw-Hill Book Company.

Cobliner, W. Godfrey (1950), "Feminine Fashion as an Aspect of Group Psychology: Analysis of Written Replies Received by Means of a Questionnaire," *Journal of Social Psychology*, 31 (March), 283–9.

Clum, Terry L. and Joanne B. Eicher (1972), "Teenagers' Conformity in Dress and Peer Friendship Groups," Research Report 156, Michigan State University Agricultural Experiment Station.

Crane, Diana (1969), "Fashion in Science: Does It Exist?" *Social Problems*, 16 (Spring), 433–41.

Daniels, Alfred H. (1951), "Fashion Merchandising," *Harvard Business Review*, 29 (May), 51–60.

Davis, Leslie L. and Franklin G. Miller (1983), "Conformity and Judgments of Fashionability," *Home Economics Research Journal*, 11 (June), 337–42.

Dooley, William H. (1930), *Clothing and Style*. New York: D. C. Heath and Company.

Eicher, Joanne B. and Eleanor A. Kelley (1974), "A Longitudinal Study of High School Girls' Friendship Patterns, Social Class, and Clothing," Research Report 222, Michigan State University Agricultural Experiment Station.

Evans, S. Evelyn (1964), "Motivations Underlying Clothing Selection and Wearing," *Journal of Home Economics*, 56 (September), 739–43.

Festinger, Leon, Stanley Schacter, and Kurt Back (1950), *Social Pressures in Informal Groups: A Study of Human Factors in Housing*. Stanford, CA: Stanford University Press.

Field, George A. (1970), "The Status Float Phenomenon: The Upward Diffusion of Innovation," *Business Horizons*, 13 (August), 45–52.

Flacus, Louis W. (1906), "Remarks on the Psychology of Clothes," *Pedagogical Seminary*, 13, 61–83.

Frings, Gini S. (1991), *Fashion: From Concept to Consumer*. Englewood Cliffs, NJ: Simon & Schuster.

Fromkin, Howard L. (1973), "The Psychology of Uniqueness: Avoidance of Similarity and Seeking of Differentness," Paper No. 438, Institute for Research in the Behavioral, Economic, and Management Sciences, Purdue University.

Gandolfo, Giancarno (1980), *Economic Dynamics, Methods and Models*, 2nd revision. Amsterdam: North-Holland Publishing Company.

Gantmacher, F. R. (1959), *Applications of the Theory of Matrices*. New York: Interscience.

Granovetter, Mark and Roland Soong (1986), "Threshold Models of Interpersonal Effects in Consumer Demand," *Journal of Economic Behavior and Organization*, 7, 83–99.

Greeno, Daniel W., Montrose S. Sommers, and Jerome B. Kernan (1973), "Personality and Implicit Behavior Patterns," *Journal of Marketing Research*, 10 (February), 63–9.

Grindereng, Margaret P. (1967), "Fashion Diffusion," *Journal of Home Economics*, 59 (March), 171–4.

Gross, Ronald and Paul Osterman, eds. (1971), *Individualism: Man in Modern Society*. New York: Dell Publishing Co., Inc.

Gurel, Lois M., June C. Wilbur, and Lee Gurel (1972), "Personality Correlates of Adolescent Clothing Styles," *Journal of Home Economics*, 64 (March), 42–7.

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

Hicks, John R. (1950), *A Contribution to the Theory of the Trade Cycle*. London: Oxford University Press.

Hirschman, Elizabeth C. (1981), "Comprehending Symbolic Consumption: Three Theoretical Issues," in *Symbolic Consumer Behavior*, Elizabeth Hirschman and Morris B. Holbrook, eds. Ann Arbor, MI: Association for Consumer Research.

——— and Morris B. Holbrook (1981), *Symbolic Consumer Behavior*. Ann Arbor, MI: Association for Consumer Research.

Horn, Marilyn J. and Lois M. Gurel (1981), *The Second Skin*, 3rd ed. Boston: Houghton Mifflin Company.

Iannaccone, Laurence R. (1989), "Bandwagons and the Threat of Chaos: Interpersonal Effects Revisited," *Journal of Economic Behavior and Organization*, 11, 431–42.

Jacobson, Wilhelmina E. (1936), "Human Motives Underlying Fashion Changes," *Practical Home Economics*, 14 (May), 230–41.

Johnson, Barbara Hunt, Richard N. Nagasawa, and Kathleen Peters (1977), "Clothing Style Differences: Their Effect on the Impression of Sociability," *Home Economics Research Journal*, 6 (September), 58–63.

King, Charles W. (1963), "Fashion Adoption: A Rebuttal to the 'Trickle-Down' Theory," in *Toward Scientific Marketing*, Stephen A. Greyser, ed. Chicago: American Marketing Association, 108–25.

——— and John O. Summers (1967), "Interaction Patterns in Interpersonal Communication," Paper No. 425, Institute for Research in the Behavioral, Economic, and Management Sciences, Purdue University.

Klapp, Orin (1969), *Collective Search for Identity*. New York: Holt, Rinehart and Winston, Inc.

Kroeber, A. L. (1919), "On the Principle of Order in Civilization as Exemplified by Changes of Fashion," *American Anthropologist*, 21, 235–63.

Lang, Kurt and Gladys Lang (1961), *Collective Dynamics*. New York: Thomas Y. Crowell.

Leibenstein, H. (1950), "Bandwagon, Snob and Veblen Effects in the Theory of Consumers' Demand," *Quarterly Journal of Economics*, 64, 183–207.

Levy, Sidney (1959), "Symbols for Sale," *Harvard Business Review*, 37 (July/August), 117–24.

Mahajan, Vijay, Eitan Muller, and Frank M. Bass (1990), "New Product Diffusion Models in Marketing: A Review and Directions for Resarch," *Journal of Marketing*, 54 (January), 1–25.

McAlister, Leigh and Edgar Pessemier (1982), "Variety Seeking Behavior: An Interdisciplinary Review," *Journal of Consumer Research*, 9 (December), 311–22.

McIntyre, Shelby H. and Christopher M. Miller (1992), "Social Utility and Fashion Behavior," *Marketing Letters*, 3 (4), 371–82.

Meyersohn, Rolf and Elihu Katz (1957), "Notes on a Natural History of Fads," *American Journal of Sociology*, 62 (May), 594–601.

Midgley, David G. (1983), "Patterns of Interpersonal Information Seeking for the Purchase of a Symbolic Product," *Journal of Marketing Research*, 20 (February), 74–83.

Norton, John A. and Frank M. Bass (1987), "A Diffusion Theory Model of Adoption and Substitution for Successive Generations of High Technology Products," *Management Science*, 33 (September), 1069–86.

Nystrom, Paul H. (1928), *Economics of Fashion*. New York: Ronald Press.

Reynolds, William H. (1968), "Cars and Clothing: Understanding Fashion Trends," *Journal of Marketing*, 32 (July), 44–9.

Richardson, Jane and A. L. Kroeber (1940), "Three Centuries of Women's Dress Fashions, A Quantitative Analysis," *Anthropological Records*, 5, 111–53.

Robinson, Dwight E. (1958), "Fashion Theory and Product Design," *Harvard Business Review*, 36 (November/December), 126–38.

——— (1961), "The Economics of Fashion Demand," *Quarterly Journal of Economics*, 75 (August), 376–98.

——— (1975), "Style Changes: Cyclical, Inexorable, and Foreseeable," *Harvard Business Review*, 53 (November/December), 121–31.

——— (1976), "Fashions in Shaving and Trimming of the Beard: The Men of the Illustrated London News, 1842–1972," *American Journal of Sociology*, 81 (March), 1133–41.

Sapir, Edward (1931), "Fashion," in *Encyclopedia of the Social Sciences*, Vol. 6. New York: Macmillan Publishing Company.

Schrank, Holly L. and D. Lois Gilmore (1973), "Correlates of Fashion Leadership: Implications for Fashion Process Theory," *Sociological Quarterly*, 14, 534–43.

Shepard, D. (1973), "Fashions in Christian Names," in *Fashion Marketing: An Anthology of Viewpoints and Perspectives*, Gordon Wills and David Midgley, eds. London: George Allen & Unwin, Ltd.

Simmel, Georg (1904), "Fashion," *International Quarterly*. Reprinted in *American Journal of Sociology*, 62 (May 1957), 541–58.

Sirgy, M. Joseph (1982), "Self-Concept in Consumer Behavior: A Critical Review," *Journal of Consumer Research*, 9 (December), 287–300.

Smucker, Betty and Anna M. Creekmore (1972), "Adolescents' Clothing Conformity, Awareness, and Peer Acceptance," *Home Economics Research Journal*, 1 (March), 92–7.

Solomon, Michael R. (1983), "The Role of Products as Social Stimuli: A Symbolic Interactionist Approach," *Journal of Consumer Research*, 10 (December), 319–29.

Sproles, George B. (1979), *Fashion: Consumer Behavior Toward Dress*. Minneapolis: Burgess.

——— (1981), "Analyzing Fashion Life Cycles—Principles and Perspectives," *Journal of Marketing*, 45 (Fall), 116–24.

——— (1985), "Behavioral Science Theories of Fashion," in *The Psychology of Fashion*, Michael R. Solomon, ed. Lexington, MA: Lexington Books.

Stigler, George and Gary S. Becker (1977), "De Gustibus Non Est Disputandum," *American Economic Review*, 67 (2), 76–90.

Summers, John D. (1970), "The Identity of Women's Clothing Fashion Opinion Leaders," *Journal of Marketing Research*, 7 (May), 117–85.

Taylor, James W. (1974), "The Role of Risk in Consumer Behavior," *Journal of Marketing*, 38 (April), 54–60.

Veblen, Thorstein (1912), *The Theory of the Leisure Class*. New York: Macmillan Co.

Venkatesan, M. (1966), "Experimental Study of Consumer Behavior Conformity and Independence," *Journal of Marketing Research*, 3 (November), 384–7.

Wasson, Chester R. (1968), "How Predictable Are Fashion and Other Product Life Cycles?" *Journal of Marketing*, 32 (July), 36–43.

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms

Weeden, Patricia (1977), "Study Patterned on Kroeber's Investigation of Style," *Dress—Journal of the Costume Society of America*, 3, 9–19.

Winakor, Geitel, Bernetta Canton, and Leroy Wolins (1980), "Perceived Fashion Risk and Self-Esteem of Males and Females," *Home Economics Research Journal*, 9 (September), 45–56.

Young, Agnes Brooks (1937), *Recurring Cycles of Fashion, 1760–1937*. New York: Harper and Brothers.

Zajonc, Robert B. and Hazel Markus (1982), "Affective and Cognitive Factors in Preferences," *Journal of Consumer Research*, 9 (September), 123–31.

Zaltman, Gerald, Karen LeMasters, and Michael Heffring (1982), *Theory Construction in Marketing*. New York: John Wiley & Sons, Inc.

——— and Melanie Wallendorf (1979), *Consumer Behavior: Basic Findings and Management Implications*. New York: John Wiley & Sons, Inc.

Reprint No. JMR302101

AMERICAN MARKETING ASSOCIATION

REPRINT AND PERMISSION POLICIES AND PROCEDURES FOR JOURNAL ARTICLES

The American Marketing Association is eager to serve you in the sharing of the valuable information found in AMA's journal articles.

REPRINTS

If you wish to order **REPRINTS** of any article, write to the American Marketing Association Publications Group, 250 S. Wacker Dr., Chicago, IL 60606 USA. Call (312) 993-9517 or Fax (312) 993-7540.

The prices below apply to *each title ordered.* Multiple titles may be combined to reach the minimum order amount of $15. Each reproduction will be provided with an attached cover. **ORDERS INCLUDING SHIPPING AND HANDLING CHARGES MUST BE PREPAID by check (in U.S. funds or equivalent) or major credit card (VISA, American Express, MasterCard).** No refunds or exchanges.

REPRINT PRICES (minimum order is $15)

1-5*	reproductions of the same article or an assortment of articles	$7.50 each	+	$6	for U.S. Shipping & Handling	or	$9	for International Air S & H
6-14*	"	$6 "	+	$9	"	or	$14	"
15-49	reproductions of the same article	$5 "	+	$12	"	or	$21	"
50-149	"	$3 "	+	$19	"	or	$42	"
150-500	"	$2 "	+	$34	"	or	$87	"
> 500	"	call for quote						

* Please add $5 per title for articles more than 15 years old.

An order of less than 50 copies also may be obtained from University Microfilms International, 300 N. Zeeb Road, Ann Arbor, MI 48106. Articles are priced prepaid at $10.75 plus $2.25 for each additional copy of the same article. Those who have a standing account with UMI pay $8.75 per article plus $2.25 for each additional copy of the same article.

PERMISSIONS

AMA would be proud to grant permission for the reproduction of its copyrighted materials. To obtain permission to reproduce multiple copies, contact the AMA Publications Group (312) 831-2751. Under the "fair use" provision of the U.S. Copyright Law, anyone may make **a photocopy** for his or her own use without seeking permission.

To secure permission to reproduce one's own works in quantity, please contact the AMA Publications Group.

AMERICAN MARKETING ASSOCIATION

This content downloaded from 68.195.21.30 on Tue, 02 Jan 2018 02:01:49 UTC
All use subject to http://about.jstor.org/terms